

Fundamentals of AI - Chapter-wise Important Questions

Unit 1: Introduction (30%)

Conceptual & Short Questions

- Define Artificial Intelligence. How is it different from traditional programming?
- Differentiate between Artificial Narrow Intelligence, General Intelligence, and Super Intelligence.
- What are agents and environments in AI?
- Explain the concept of rationality in intelligent agents.

Descriptive Questions

- Discuss the history and evolution of Artificial Intelligence.
- Explain the characteristics of intelligent agents with suitable examples.
- Describe different types of environments in AI.
- What are production systems in AI? Explain with an example.

Unit 2: Knowledge Representation (20%)

Conceptual & Short Questions

- What is knowledge representation in AI? Why is it important?
- Define Natural Language Processing (NLP).
- What is pattern recognition? Give one example.
- What are expert systems? Mention any two examples.

Descriptive Questions

- Explain various approaches to knowledge representation in AI.
- Discuss the role of NLP in AI applications.
- How do expert systems work? Describe their components.

Unit 3: Basics of Machine Learning (20%)

Conceptual & Short Questions

- Define supervised, unsupervised, and reinforcement learning with examples.
- What is the difference between linear and logistic regression?
- What are the applications of Support Vector Machines (SVM)?

Descriptive Questions

- Explain the concept of learning from examples with appropriate ML models.
- Compare and contrast supervised and unsupervised learning.
- Describe the working of a Support Vector Machine with a simple example.

Unit 4: Deep Learning (20%)

Conceptual & Short Questions

- What is deep learning? How is it different from traditional ML?
- Define and explain neural networks.
- List types of deep learning models.

Descriptive Questions

- Describe the architecture of a neural network.
- Explain the applications of deep learning in real life.
- Discuss types of deep learning models with examples.

Unit 5: Modern Artificial Intelligence (10%)

Conceptual & Short Questions

- What are Large Language Models (LLMs)?
- Mention any two use-cases of LLMs like ChatGPT, Gemini, etc.
- What are the current issues faced by AI?

Descriptive Questions

- Explain the working of any one LLM (e.g., ChatGPT).
- What are the future challenges in the development of AI?
- Discuss the impact of modern AI systems on society.